

Practice Quiz: Learning (Computational Neuroscience)

Name: __ Date: __

Part A: Multiple Choice

1. Hebbian learning states that: A) Neurons that fire together weaken their connections B) Neurons that fire together strengthen their connections (wire together) C) Learning only occurs during sleep D) Only motor neurons can learn
2. NMDA receptors act as coincidence detectors because: A) They only respond to one neurotransmitter B) They require both presynaptic glutamate and postsynaptic depolarization to open C) They detect sounds D) They are only found in the hippocampus
3. Dopamine neurons fire a burst when: A) An expected reward is received on time B) An unexpected reward is received (positive reward prediction error) C) Nothing happens D) The animal is sleeping
4. During sleep, the hippocampus: A) Shuts down completely B) Replays recorded experiences during sharp-wave ripples for cortical consolidation C) Only processes emotions D) Deletes all memories from the day
5. STDP strengthens a synapse when: A) The postsynaptic neuron fires first B) The presynaptic neuron fires just before the postsynaptic neuron C) Both neurons fire at exactly the same time D) Neither neuron fires
6. Place cells in the hippocampus: A) Store memories of faces B) Fire when an animal is at specific locations, forming a cognitive map C) Only exist in humans D) Control muscle movement
7. Addiction may involve dopamine-based learning gone wrong because: A) Drugs have no effect on dopamine B) Drugs generate artificially large reward prediction errors, strongly

reinforcing drug-seeking behavior C) Addicts have damaged hippocampi D) Dopamine is not involved in learning

Part B: Short Answer

1. Explain how the NMDA receptor implements a form of Bayesian updating at the molecular level. What represents the "prior" and what represents the "evidence" in this analogy?
2. A rat learns that pressing a lever produces food. Using dopamine prediction errors, trace what happens to the dopamine signal (a) when the rat first discovers this, (b) after many successful presses, and (c) when the food suddenly stops coming.
3. Compare hippocampal learning and cerebellar learning. How do they differ in what they learn, how they learn it, and what types of memory they support?